CampusConnect 2026

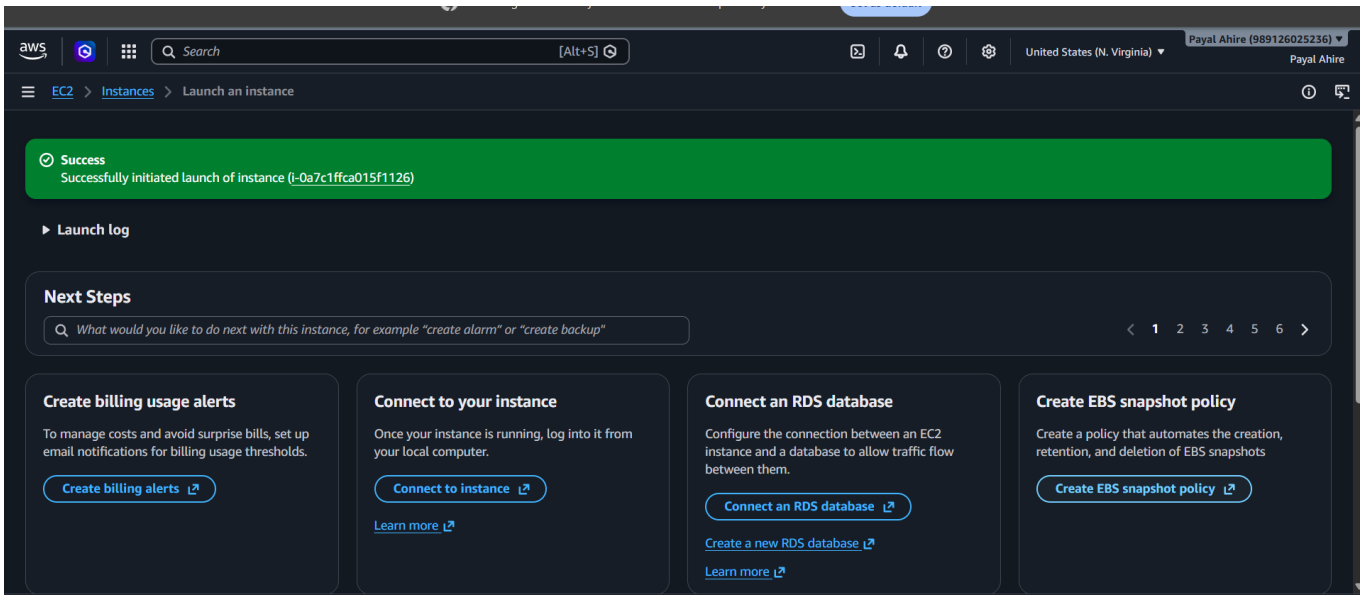
AWS EC2 – Nginx – MariaDB – PHP
Practical Screenshot & Short Explanation Report

Project	CampusConnect 2026
Cloud Platform	AWS EC2
Web Server	Nginx
Database	MariaDB
Backend	PHP
Evidence	Screenshots arranged in the required 1–15 sequence

All available matching screenshots from the supplied chat evidence have been used; no step is intentionally left empty.

1. EC2 Instance Launch

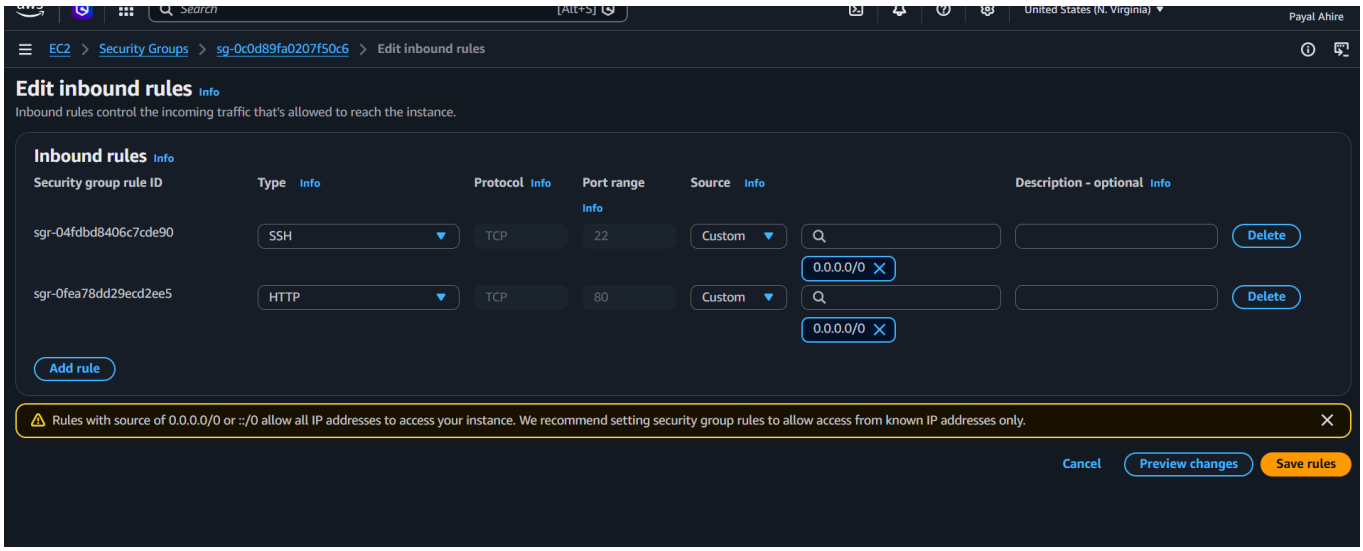
The AWS EC2 instance was successfully launched. The screenshot shows the successful launch message and the instance ID.



Screenshot evidence from the supplied practical work.

2. Security Group – HTTP and SSH

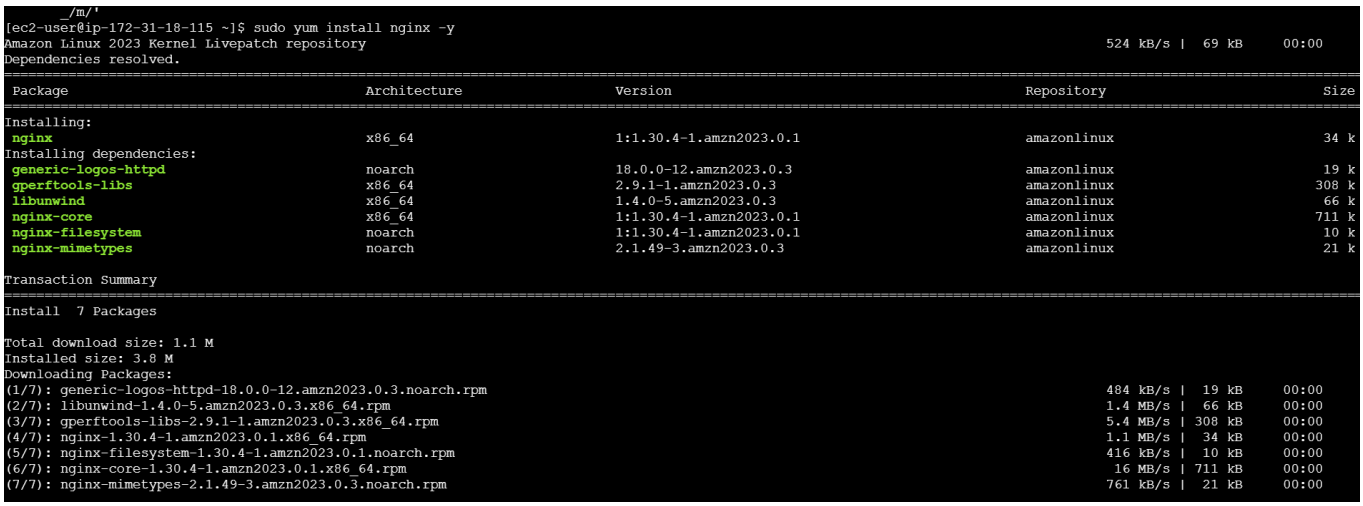
The Security Group allows SSH on port 22 for remote administration and HTTP on port 80 so users can access the web application through a browser.



Screenshot evidence from the supplied practical work.

3. Install Nginx

Nginx was installed on Amazon Linux using yum. Nginx works as the web server and handles incoming HTTP requests.



```
       _/m/'
[ec2-user@ip-172-31-18-115 ~]$ sudo yum install nginx -y
Amazon Linux 2023 Kernel Livepatch repository                                                    524 kB/s |  69 kB     00:00
Dependencies resolved.
================================================================================================================================
 Package                                 Architecture                Version                                 Repository                          Size
================================================================================================================================
Installing:
 nginx                                   x86_64                      1:1.30.4-1.amzn2023.0.1                 amazonlinux                         34 k
Installing dependencies:
 generic-logos-httpd                     noarch                      18.0.0-12.amzn2023.0.3                  amazonlinux                         19 k
 gperftools-libs                         x86_64                      2.9.1-1.amzn2023.0.3                    amazonlinux                        308 k
 libunwind                               x86_64                      1.4.0-5.amzn2023.0.3                    amazonlinux                         66 k
 nginx-core                              x86_64                      1:1.30.4-1.amzn2023.0.1                 amazonlinux                        711 k
 nginx-filesystem                        noarch                      1:1.30.4-1.amzn2023.0.1                 amazonlinux                         10 k
 nginx-mimetypes                         noarch                      2.1.49-3.amzn2023.0.3                   amazonlinux                         21 k

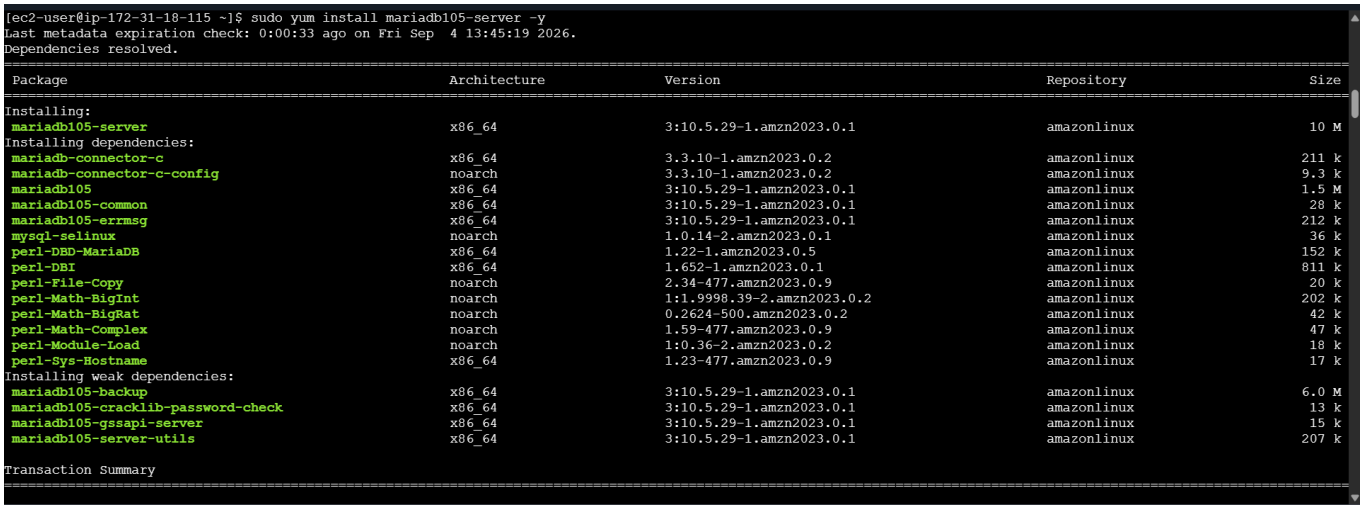
Transaction Summary
================================================================================================================================
Install  7 Packages

Total download size: 1.1 M
Installed size: 3.8 M
Downloading Packages:
(1/7): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm                                     484 kB/s |  19 kB     00:00
(2/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm                                                 1.4 MB/s |  66 kB     00:00
(3/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm                                           5.4 MB/s | 308 kB     00:00
(4/7): nginx-1.30.4-1.amzn2023.0.1.x86_64.rpm                                                    1.1 MB/s |  34 kB     00:00
(5/7): nginx-filesystem-1.30.4-1.amzn2023.0.1.noarch.rpm                                         416 kB/s |  10 kB     00:00
(6/7): nginx-core-1.30.4-1.amzn2023.0.1.x86_64.rpm                                                16 MB/s | 711 kB     00:00
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm                                          761 kB/s |  21 kB     00:00
```

Screenshot evidence from the supplied practical work.

4. Install MariaDB

MariaDB Server was installed using yum. MariaDB is used to store and retrieve the student registration data.



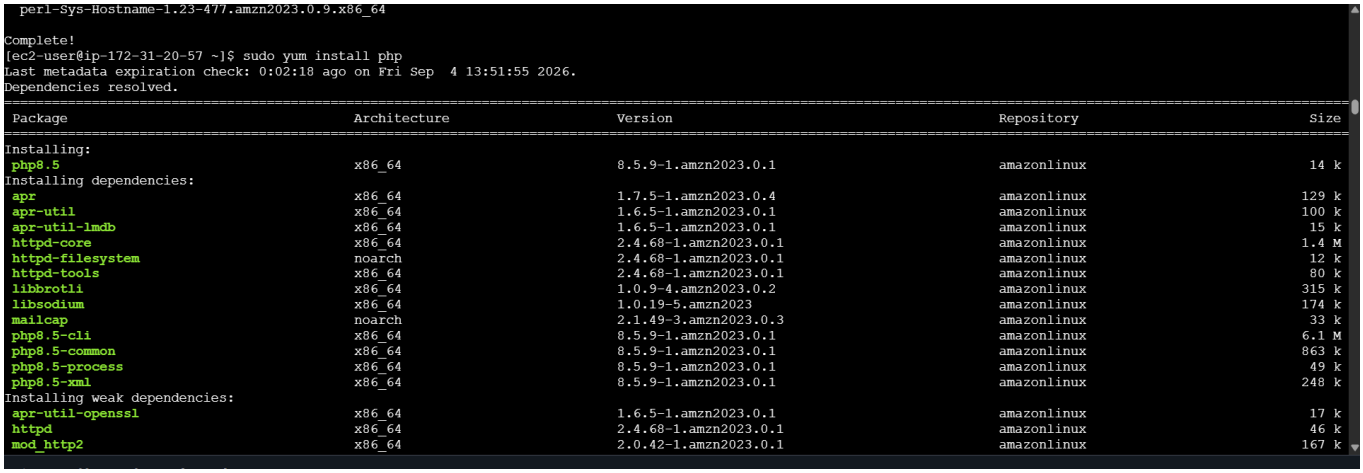
```
[ec2-user@ip-172-31-18-115 ~]$ sudo yum install mariadb105-server -y
Last metadata expiration check: 0:00:33 ago on Fri Sep  4 13:45:19 2026.
Dependencies resolved.
================================================================================
 Package                    Architecture  Version                  Repository  Size
================================================================================
Installing:
 mariadb105-server          x86_64        3:10.5.29-1.amzn2023.0.1 amazonlinux 10 M
Installing dependencies:
 mariadb-connector-c        x86_64        3.3.10-1.amzn2023.0.2    amazonlinux 211 k
 mariadb-connector-c-config noarch        3.3.10-1.amzn2023.0.2    amazonlinux 9.3 k
 mariadb105                 x86_64        3:10.5.29-1.amzn2023.0.1 amazonlinux 1.5 M
 mariadb105-common          x86_64        3:10.5.29-1.amzn2023.0.1 amazonlinux 28 k
 mariadb105-errmsg          x86_64        3:10.5.29-1.amzn2023.0.1 amazonlinux 212 k
 mysql-selinux              noarch        1.0.14-2.amzn2023.0.1    amazonlinux 36 k
 perl-DBD-MariaDB           x86_64        1.22-1.amzn2023.0.5      amazonlinux 152 k
 perl-DBI                   x86_64        1.652-1.amzn2023.0.1     amazonlinux 811 k
 perl-File-Copy             noarch        2.34-477.amzn2023.0.9    amazonlinux 20 k
 perl-Math-BigInt           noarch        1:1.9998.39-2.amzn2023.0.2 amazonlinux 202 k
 perl-Math-BigRat           noarch        0.2624-500.amzn2023.0.2  amazonlinux 42 k
 perl-Math-Complex          noarch        1.59-477.amzn2023.0.9    amazonlinux 47 k
 perl-Module-Load           noarch        1:0.36-2.amzn2023.0.2    amazonlinux 18 k
 perl-Sys-Hostname          x86_64        1.23-477.amzn2023.0.9    amazonlinux 17 k
Installing weak dependencies:
 mariadb105-backup          x86_64        3:10.5.29-1.amzn2023.0.1 amazonlinux 6.0 M
 mariadb105-cracklib-password-check x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 13 k
 mariadb105-gssapi-server   x86_64        3:10.5.29-1.amzn2023.0.1 amazonlinux 15 k
 mariadb105-server-utils    x86_64        3:10.5.29-1.amzn2023.0.1 amazonlinux 207 k

Transaction Summary
================================================================================
```

Screenshot evidence from the supplied practical work.

5. Install PHP

PHP was installed on the EC2 server. PHP is used for the server-side registration and login processing.



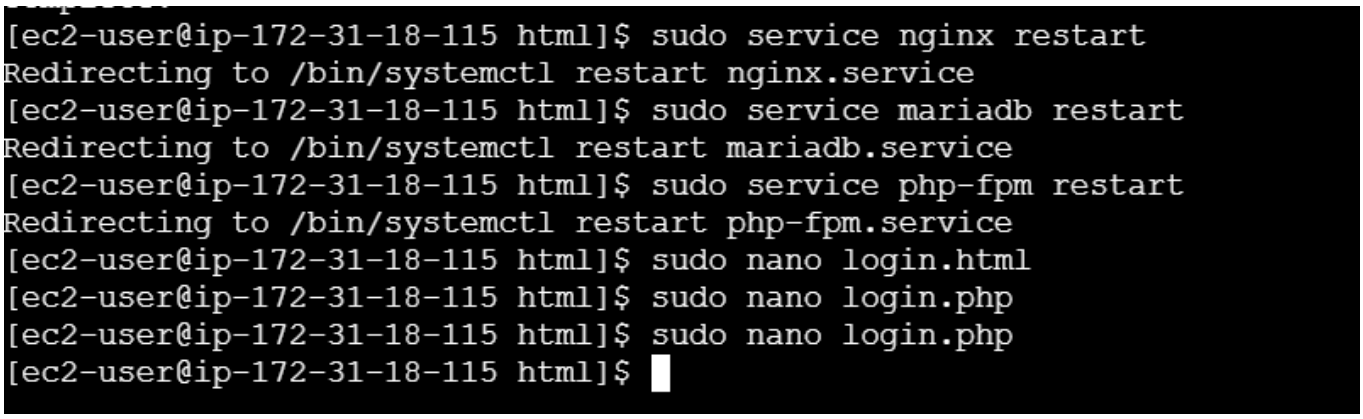
```
  perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64

Complete!
[ec2-user@ip-172-31-20-57 ~]$ sudo yum install php
Last metadata expiration check: 0:02:18 ago on Fri Sep  4 13:51:55 2026.
Dependencies resolved.
================================================================================
 Package                 Architecture    Version                 Repository    Size
================================================================================
Installing:
 php8.5                  x86_64          8.5.9-1.amzn2023.0.1    amazonlinux   14 k
Installing dependencies:
 apr                     x86_64          1.7.5-1.amzn2023.0.4    amazonlinux  129 k
 apr-util                x86_64          1.6.5-1.amzn2023.0.1    amazonlinux  100 k
 apr-util-lmdb           x86_64          1.6.5-1.amzn2023.0.1    amazonlinux   15 k
 httpd-core              x86_64          2.4.68-1.amzn2023.0.1   amazonlinux  1.4 M
 httpd-filesystem        noarch          2.4.68-1.amzn2023.0.1   amazonlinux   12 k
 httpd-tools             x86_64          2.4.68-1.amzn2023.0.1   amazonlinux   80 k
 libbrotli               x86_64          1.0.9-4.amzn2023.0.2    amazonlinux  315 k
 libsodium               x86_64          1.0.19-5.amzn2023       amazonlinux  174 k
 mailcap                 noarch          2.1.49-3.amzn2023.0.3   amazonlinux   33 k
 php8.5-cli              x86_64          8.5.9-1.amzn2023.0.1    amazonlinux  6.1 M
 php8.5-common           x86_64          8.5.9-1.amzn2023.0.1    amazonlinux  863 k
 php8.5-process          x86_64          8.5.9-1.amzn2023.0.1    amazonlinux   49 k
 php8.5-xml              x86_64          8.5.9-1.amzn2023.0.1    amazonlinux  248 k
Installing weak dependencies:
 apr-util-openssl        x86_64          1.6.5-1.amzn2023.0.1    amazonlinux   17 k
 httpd                   x86_64          2.4.68-1.amzn2023.0.1   amazonlinux   46 k
 mod_http2               x86_64          2.0.42-1.amzn2023.0.1   amazonlinux  167 k
```

Screenshot evidence from the supplied practical work.

6. Start All Services

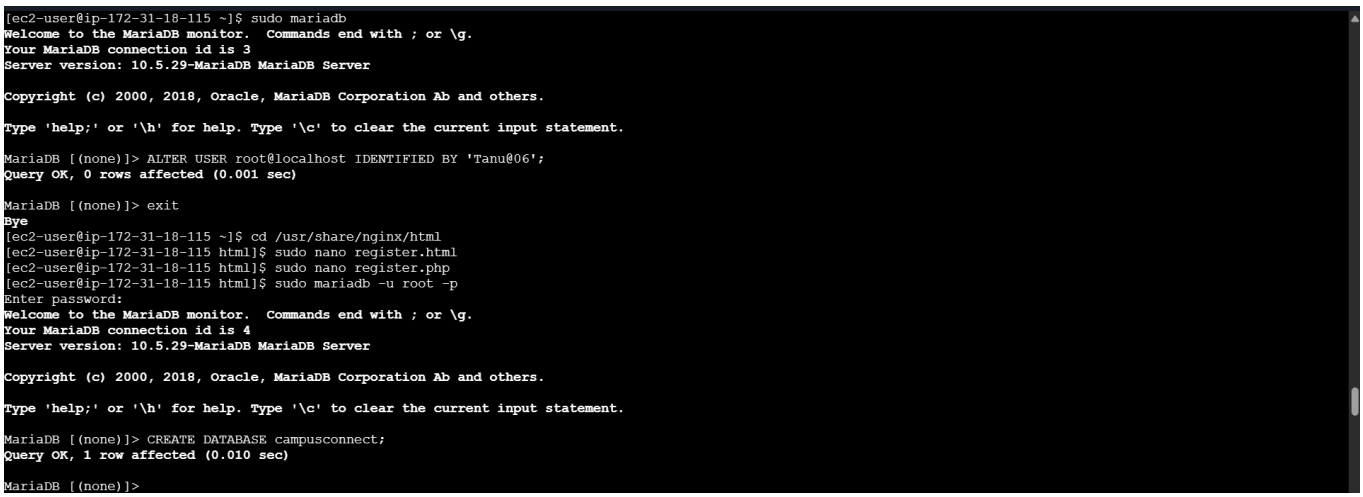
Nginx, MariaDB and PHP-FPM services were restarted so that the web server, database and PHP processing environment could work together.



Screenshot evidence from the supplied practical work.

7. Create Database

The MariaDB database named campusconnect was created successfully using CREATE DATABASE campusconnect;.



Screenshot evidence from the supplied practical work.

8. Create Student Table

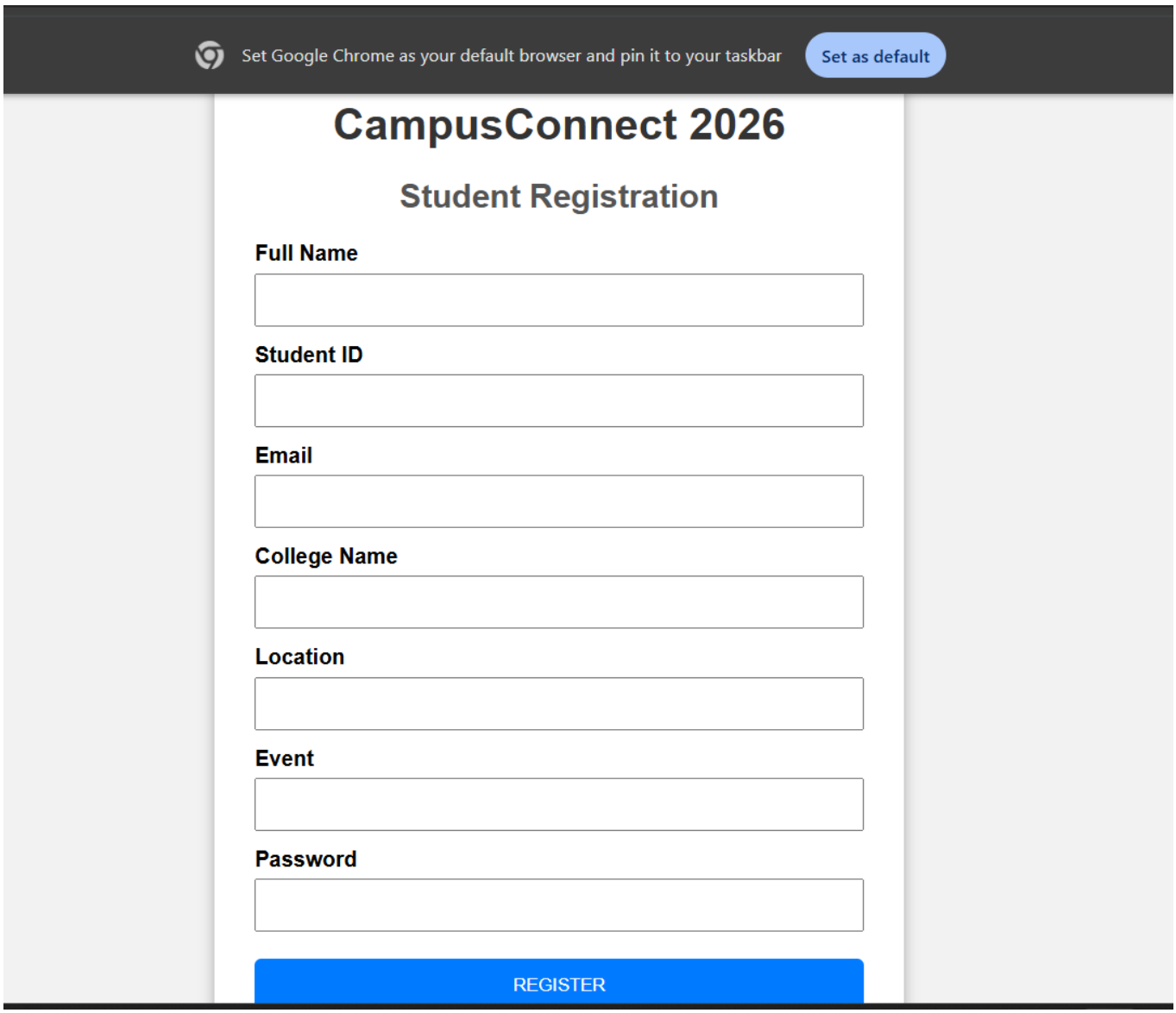
The student1 table was created inside the campusconnect database. It contains fields for ID, fullname, student ID, email, college, location, event and password.



Screenshot evidence from the supplied practical work.

9. Create register.html

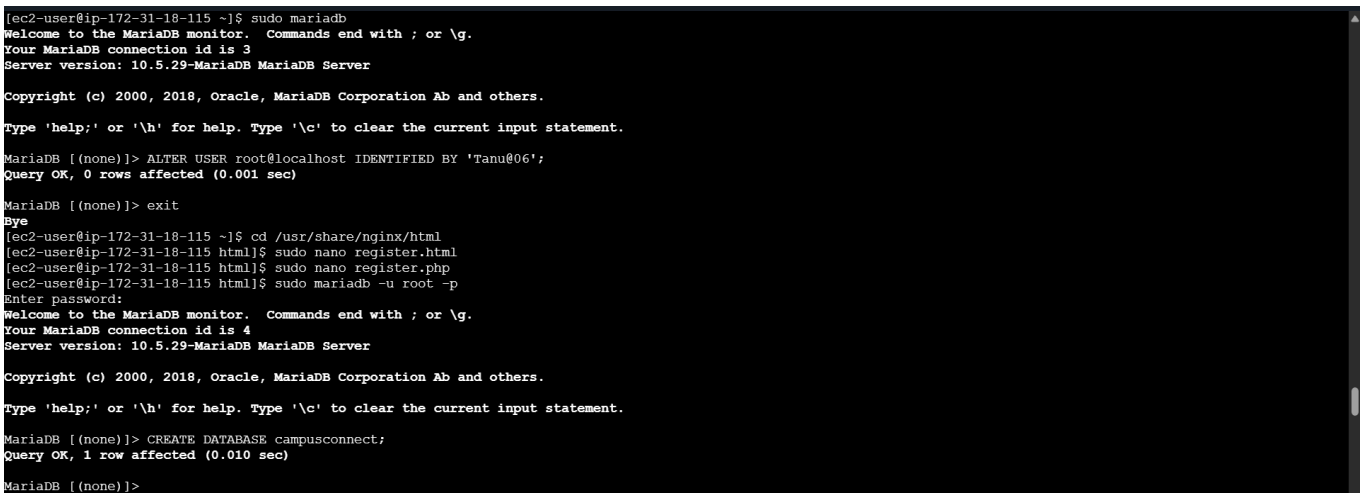
The registration page was created with the required fields: Full Name, Student ID, Email, College Name, Location, Event and Password, with a Register button.



Screenshot evidence from the supplied practical work.

10. Create register.php

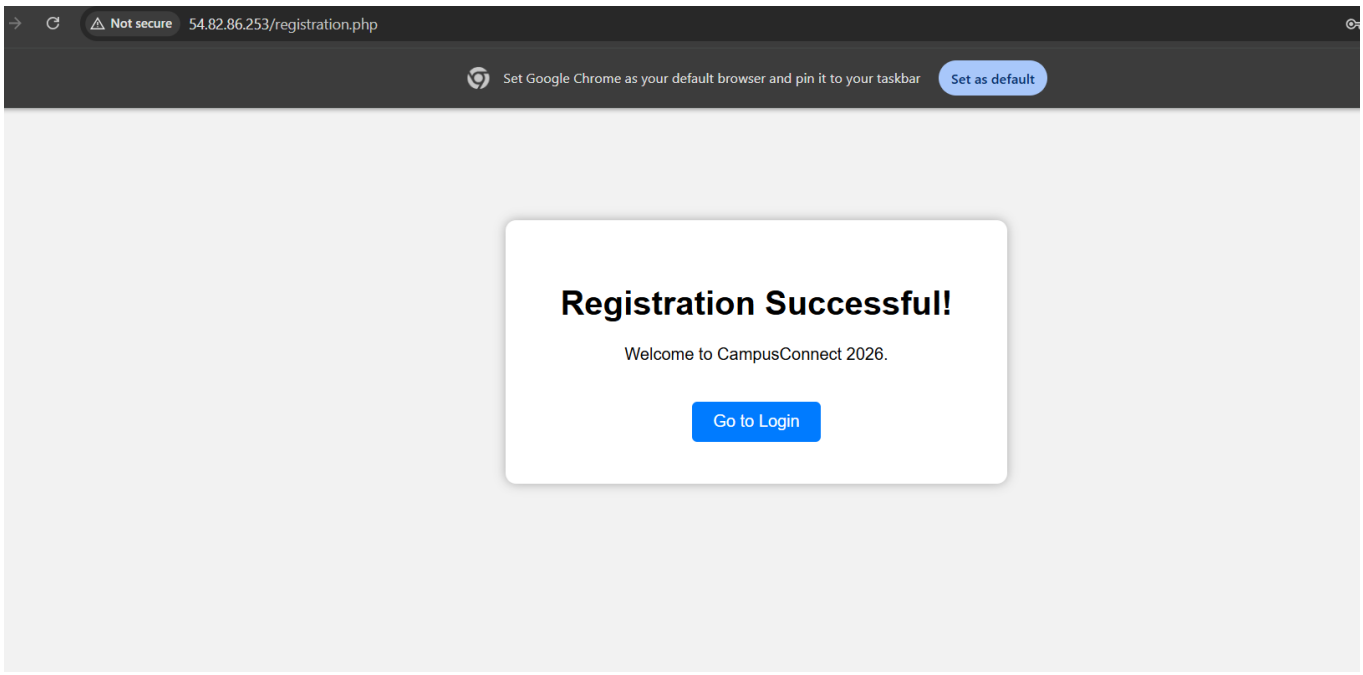
The register.php file was created in the Nginx web directory. It is used to process registration data submitted from the registration form.



Screenshot evidence from the supplied practical work.

11. Registration Page Output

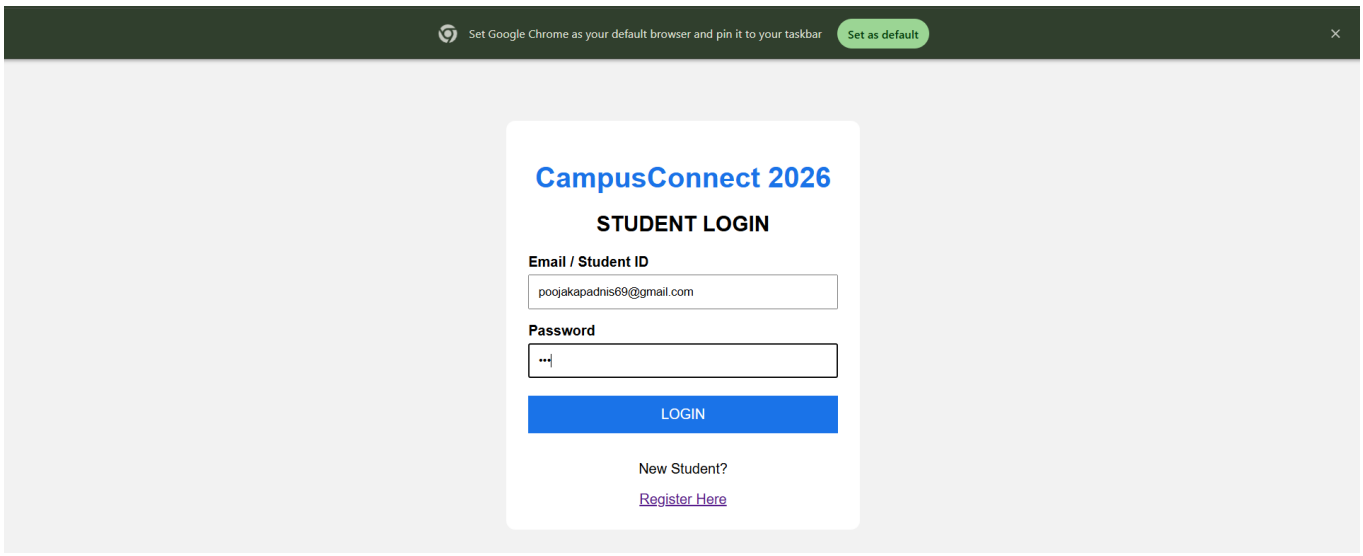
After successful form submission, the website displayed 'Registration Successful!' and provided a button to continue to Login.



Screenshot evidence from the supplied practical work.

12. Create login.html

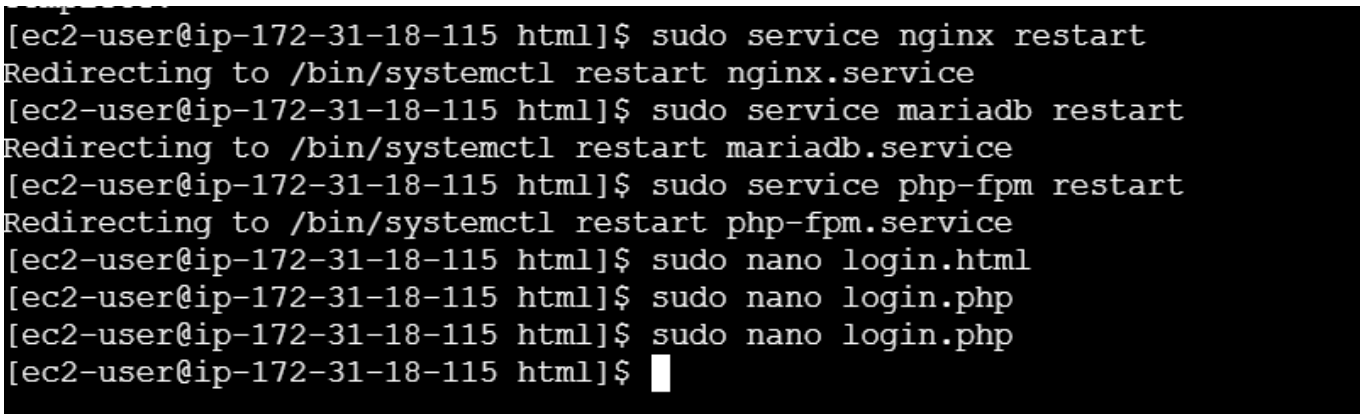
The login page contains an Email / Student ID field, Password field, Login button and a link for new students to register.



Screenshot evidence from the supplied practical work.

13. Create login.php

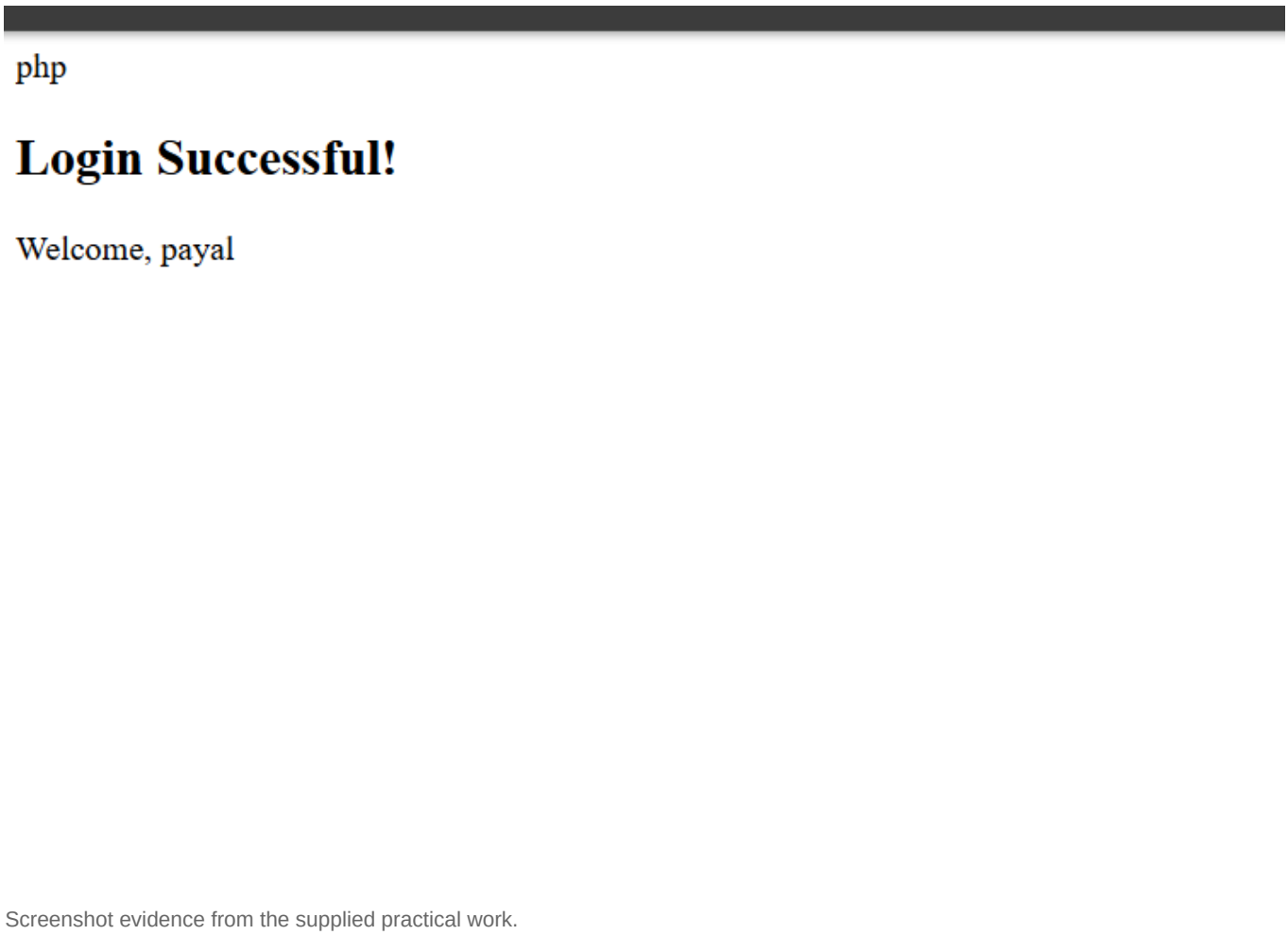
The login.php file was created for processing login credentials. It checks the submitted Email/Student ID and password against the registered student data.



Screenshot evidence from the supplied practical work.

14. Login Successful

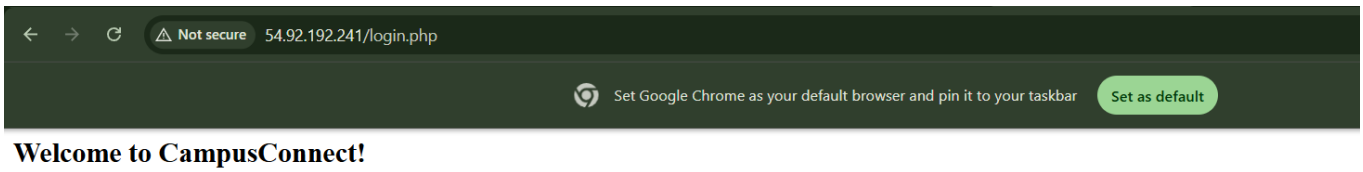
The login operation was successful and the application displayed 'Login Successful!' with a welcome message for the logged-in student.



Screenshot evidence from the supplied practical work.

15. Final Website Through Server IP

The final website was accessed through the EC2 public IP address. The browser address bar shows the server IP, confirming that the application is deployed and accessible through the server.



Screenshot evidence from the supplied practical work.